

Silica in Seawater

1 SCOPE

This method covers the determination of Silica in seawater.

2 RANGE OF APPLICATION

Range 0.1 – 10.0 mg silica/L

3 METHOD DETECTION LIMIT

By EPA Procedure MDL = 0.011 mg silica/L

4 METHOD PRINCIPLE

Under acidic conditions, molybdate-reactive silica combines with ammonium molybdate to form a yellow molybdo-silicic acid complex. This complex is chemically reduced by adding buffered 4-amino-3-hydroxy-1-naphthalenesulfonic acid to form a silico-molybdenum blue complex. The absorbance of the final product is measured spectrophotometrically at 660 nm.

5 REFERENCES

Methods for Chemical Analysis of Waters and Wastes, US EPA 600/4-79-020, 1983: Method 370.1
Standards Methods for the Examination of Water and Wastewater, APHA/AWWA/WEF, method 4500 SiO₂ – D (2000 forward).

6 INTERFERENCES

Inclusion of oxalic acid in the reagent system minimizes any possible interference by phosphorus and eliminates tannin interference.

Large amounts of iron or sulfide interfere, as do sample color and turbidity.

Bias from sample turbidity or color is correctable using sample blanking feature of AQ software.

7 REAGENTS: CHEMICALS REQUIRED AND SAFETY INFORMATION

Consult the safety data sheets (SDS) for details on safe handling of chemicals.

<u>Chemical</u>	<u>Formula</u>	<u>C.A.S. No.</u>	<u>Safety Classification</u>
4-amino-3-hydroxy-1-naphthalenesulfonic acid	$\text{H}_{10}\text{C}_{10}\text{O}_4\text{SN}$	116-63-2	--
Ammonium molybdate tetrahydrate	$(\text{NH}_4)_6\text{Mo}_7\text{O}_{24} \cdot 4\text{H}_2\text{O}$	12054-85-2	--
Hydrochloric acid	HCl	7647-01-0	GHS05, GHS07
Oxalic acid dihydrate	$\text{C}_2\text{H}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$	6153-56-6	GHS05, GHS07
Sodium bicarbonate*	NaHCO_3	144-55-8	--
Sodium bisulfite	NaHSO_3	7631-90-5	GHS05, GHS07
Sodium chloride*	NaCl	7647-14-5	--
Sodium hydroxide	NaOH	1310-73-2	GHS05
Sodium metasilicate nonahydrate	$\text{Na}_2\text{O}_3\text{Si} \cdot 9\text{H}_2\text{O}$	13517-24-3	GHS05, GHS07
Sodium sulfite	Na_2SO_3	7757-83-7	--
GHS01: Danger – Explosive	GHS02: Danger – Flammable	GHS03: Danger – Oxidizing	
GHS04: Warning – Compressed Gas	GHS05: Warning/Danger – Corrosive	GHS06: Danger – Toxic	
GHS07: Warning – Irritant	GHS08: Danger – Health Hazard	GHS09: Warning/Danger – Aquatic Toxicity	

Note: Persons using this method should be familiar with normal laboratory practice. This method does not purport to address all of the safety problems, if any, associated with its use. It is the responsibility of the user to establish appropriate safety and health practices and to ensure compliance with local regulatory conditions.

* For use when making synthetic seawater only

8 REAGENT SOLUTIONS

Use reagent-grade chemicals certified for analytical or general laboratory use. SEAL recommends using ACS Reagent Grade whenever possible. Reagents of technical or commodity grade must be validated by the user. Use high-purity reagent water, distilled or deionized, and free from organic contamination, therefore, Grade 1 (ISO Standard 3696) or better than Type II (ASTM Standard D1193) waters are suitable.

8.1 OXALIC ACID SOLUTION, 10% w/v

Oxalic acid dihydrate	10 g
DI Water	dilute to 100 mL

Dissolve 10 g oxalic acid dihydrate in approximately 80 mL of deionized water in a 100 mL plastic volumetric flask. Dilute to the mark with deionized water.

Store at 4°C in a plastic bottle. Prepare fresh every third day from dry chemical.

If results are erratic, purchase fresh reagent from an alternate vendor if necessary.

8.2 HYDROCHLORIC ACID, 10% v/v

Hydrochloric acid, concentrated	10 mL
DI Water	dilute to 100 mL

Add 10 mL of hydrochloric acid, concentrated to approximately 80 mL of deionized water in a 100 mL plastic volumetric flask. **CAUTION**, the flask will become warm! Cool to room temperature and dilute to the mark with deionized water.

Store at room temperature in a plastic bottle.

8.3 AMMONIUM MOLYBDATE, 10% w/v

Ammonium molybdate	10 g
Sodium hydroxide pellets, for pH adjustment	pH 7.0 ± 0.1
DI Water	dilute to 100 mL

Place 10 g ammonium molybdate and approximately 80 mL of deionized water in a 250 mL plastic beaker. Stir with gentle warming for 30 minutes. Add 1 g of sodium hydroxide and stir until completely dissolved. Cool to room temperature and check pH of the solution. Add additional pellets of sodium hydroxide, 1 or 2 at a time while stirring, until pH 7.0 ± 0.1 is achieved. Filter if turbidity is seen. Transfer to 100 mL plastic volumetric flask, dilute to the mark with deionized water.

Store at 4°C in a plastic bottle. Stable for 1 month. Discard if reagent becomes blue or turbid.

8.4 REDUCING AGENT

4-amino-3-hydroxy-1-naphthalenesulfonic acid	500 mg
Sodium sulfite	1.0 g
Sodium bisulfite	30 g
DI Water, divided	200 mL

Dissolve 500 mg 4-amino-3-hydroxy-1-naphthalenesulfonic acid and 1.0 g sodium sulfite in 50 mL of deionized water with gentle warming. In a separate flask, dissolve 30 g sodium bisulfite in 150 mL deionized water. Mix these two solutions and filter into an opaque plastic bottle.

Store at 4°C in a plastic bottle. Stable for 1 month. Discard if solution becomes dark.

If there is incomplete solubility or immediate darkening of the 4-amino-3-hydroxy-1-naphthalenesulfonic acid solution, do not use.

9 STANDARD SOLUTIONS

9.1 SILICA STOCK STANDARD SOLUTION (1000 mg silica/L)

Sodium metasilicate nonahydrate 4.73 g
DI water dilute to 1 L

Dissolve 4.73 g sodium metasilicate nonahydrate in approximately 900 mL of recently boiled and cooled deionized water in a 1 liter plastic volumetric flask. Dilute to the mark with deionized water and store in a plastic bottle.

Store at 4°C in a plastic bottle. Stable for 1 month.

9.2 CALIBRATION DILUENT (SYNTHETIC SEAWATER)

Sodium chloride 35 g
Sodium hydrogen carbonate 0.2 g
DI water dilute to 1 L

Dissolve 35 g of sodium chloride and 0.2 g of sodium hydrogen carbonate in about 800 mL of deionized water in a 1 liter plastic volumetric flask. Dilute to 1 liter with deionized water. Store in a plastic container.

***Note:** The concentration of sodium chloride can be adjusted to match the salinity of the seawater samples.

Store at room temperature. Stable for 1 month.

9.3 AUTO CALIBRATION STANDARD – SILICA (10.0 mg silica/L)

SILICA stock standard (1000 mg silica/L, above) 2.0 mL
CALIBRATION DILUENT dilute to 200 mL

To a 200 mL plastic volumetric flask, pipet 2.0 mL SILICA stock standard (1000 mg silica/L) and dilute to the mark with CALIBRATION DILUENT.

Store at 4°C. Prepare fresh weekly.

9.4 MANUAL CALIBRATION

These are suggested concentrations for manual calibration standards.

Concentration, mg silica/L	0	0.1	0.5	1.0	2.5	5.0	7.5	10.0
Volume (mL) Auto Calibration Standard Solution diluted as indicated with Calibration Diluent	--	1.0	5.0	10	25	50	75	100
Final Volume, mL	100	100	100	100	100	100	100	100

Prepare weekly.

10 SAMPLE PRESERVATION AND STORAGE

Samples may be collected into plastic or PTFE containers.

Samples should be analyzed as soon as practical after collection. If storage of samples is unavoidable, samples should be preserved by refrigeration with a maximum holding time of 28 days.

The laboratory should consult the EPA for the most current acceptable sample storage and preservation procedures.

11 PROCEDURE

Prepare standards and reagents as described in Sections 8 and 9. The laboratory must determine standard concentrations that are most suitable to their needs for calibration within the range of application for this method. The concentrations listed within this method are only suggestions.

Samples should be filtered through 0.45 µm membrane filter. Any reagents having visible particulates should also be filtered.

Pour calibration standards, quality control solutions and samples into sample cups and reagent wedges, according to the user-defined work list.

Before running the analysis batch, check the analyzer for sufficient reagent water, fresh reaction segments, correct location of reagent wedges, and a waste collection bottle in place.

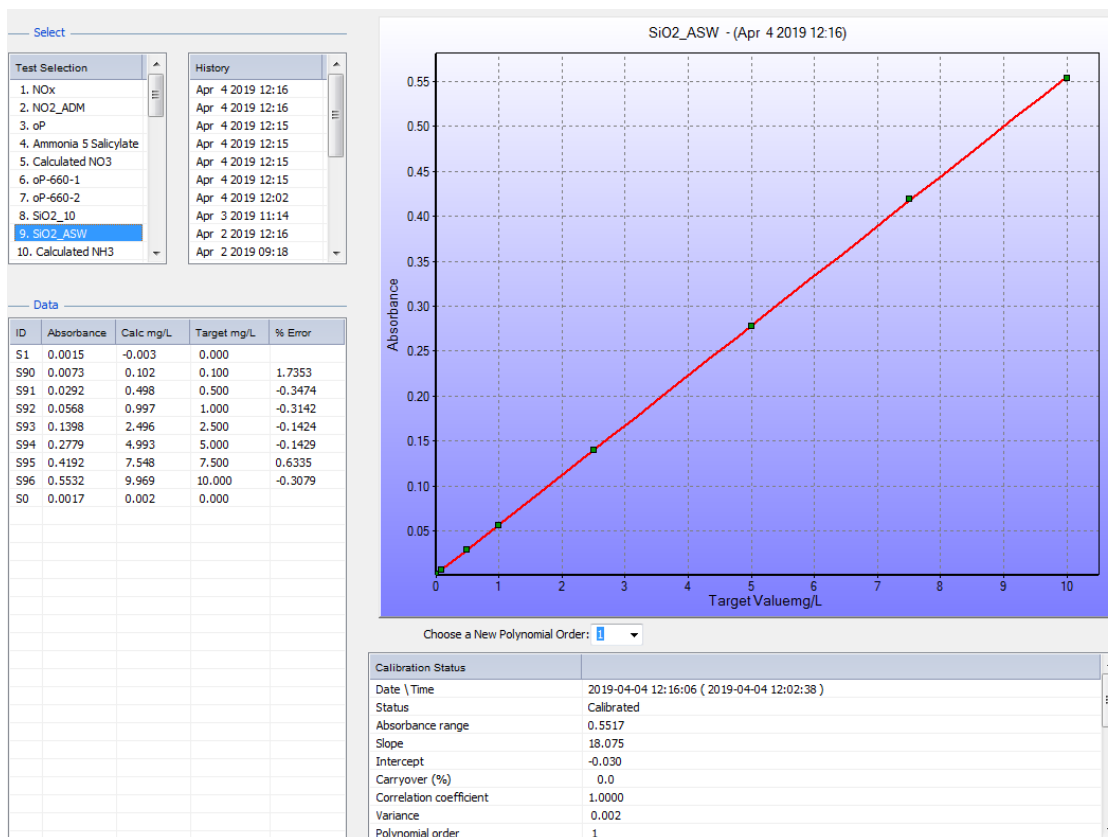
12 TEST PARAMETERS

PARAMETER	AQ SETTING
Test name	Silica SW 10
Units	mg/L
Decimals	3
Test type	End Point
Sample volume (µL)	300
Water volume (µL)	70
Number of mixes	2
Cuvette primes	5
Cuvette washes	2
Reaction time (seconds)	200
Wavelength (nm)	660
Polynomial order	1
Number of reagents	4
1. Silica – HCl (µL)	70
2. Silica – Ammonium molybdate (µL)	28
3. Silica – Oxalic acid (µL)	30
Reagent 3 Delay (sec)	110
4. Silica – Reducing reagent (µL)	20
Advanced Test Parameters	☒ Extra Debubbling Action

DEFINITIONS:

Test name:	Name which appears on the final report
Units:	Concentration units appearing against the reported result
Decimals:	Number of decimal places in the reported result
Test type:	The type of test the software will perform
Sample volume:	Volume (in microliters) of the sample used in the test
Water volume:	Volume (in microliters) of water from the water reservoir used in the test
Number of mixes:	Number of mixes of the sample + reagent in the reaction well
Cuvette primes:	Number of times the cuvette is flushed with reagent (no sample) prior to sampling
Cuvette washes:	Number of washes of the cuvette between aspirations of finished reaction liquid
Baseline on Wash:	Water baseline measured between aspirations to provide drift correction
Reaction time:	Time (in seconds) between addition of sample + reagent and measurement of the reaction mixture
Wavelength:	Interference filter wavelength to be used in the test
Polynomial order:	Polynomial order to be used when fitting the standard points to the standard graph (1 = linear)
Number of reagents:	Number of reagents defined for the test
Reagent volume:	Volume (in microliters) of the reagent used in the test

13 STANDARDIZATION



Max. Blank Absorbance (S1): 0.005 AU

Sensitivity Range (10 mg/L): 0.44 – 0.66 AU

DEFINITIONS:

Abs. Range: Difference between the absorbance of the highest standard and the absorbance of the blank

Variance: Calculated as follows:

$$\text{Variance} = \frac{\sum(\text{Deviations, i.e., errors})^2}{\sum(\text{Observed values})^2} \times 100$$

Carryover (%): Calculated as follows:

$$\text{Carryover (\%)} = \frac{(A - B)}{C} \times 100\%$$

Where, A = absorbance of carryover blank (S0)

B = absorbance of blank (S1)

C = calibration absorbance range

14 PERFORMANCE VALIDATION DATA

14.1 DETECTION LIMIT STUDY

Replicates of 0.1 mg silica/L silica standard in synthetic seawater***

Concentration (mg silica/L)				
	0.096	0.096	0.110	0.101
	0.098	0.094	0.108	0.098
	0.092	0.088	0.103	0.095
	0.105	0.096	0.111	0.092
	0.102	0.091	0.106	0.098
	0.088	0.092	0.112	0.105
	0.098	0.092	0.112	0.093
	0.093	0.090	0.111	0.100
	0.098	0.093	0.112	0.100
	0.101	0.094	0.108	0.090
	0.093	0.090	0.116	0.100
	0.096	0.097	0.105	0.099
	0.103	0.097	0.116	0.093
	0.094	0.087	0.105	0.099
	0.108	0.099	0.116	0.097
Statistics				
Mean	0.098	0.093	0.110	0.097
Std. Deviation	0.0054	0.0038	0.0043	0.0040
MDL*	0.014	0.009	0.011	0.010

14.2 PERFORMANCE STUDY

Replicates of 0.25 mg silica/L silica standard in synthetic seawater***

Concentration (mg silica/L)				
	0.234	0.238	0.262	0.246
	0.232	0.242	0.257	0.246
	0.236	0.250	0.257	0.243
	0.227	0.247	0.260	0.252
	0.236	0.248	0.258	0.240
	0.237	0.250	0.260	0.243
	0.233	0.250	0.265	0.246
	0.230	0.248	0.260	0.240
	0.236	0.246	0.259	0.247
	0.241	0.243	0.261	0.245
Statistics				
Mean	0.234	0.246	0.260	0.245
Std. Deviation	0.004	0.004	0.002	0.003
% RSD**	1.7	1.6	0.9	1.4
% Recovery	93.5	98.5	104.0	94.7

14.1 PRECISION STUDY

Replicates of 5.0 mg silica/L silica standard in synthetic seawater***

Concentration (mg silica/L)				
Statistics				
Mean	5.014	5.010	5.031	5.012
% Recovery	100.3	100.2	100.6	100.2
Std. Deviation	0.061	0.043	0.047	0.0101
% RSD**	1.2	0.9	0.9	2.0
Maximum	5.114	5.070	5.104	5.103
Minimum	4.886	4.915	4.964	4.736

Replicates of 10.0 mg silica/L silica standard in synthetic seawater***

Concentration (mg silica/L)				
Statistics				
Mean	9.959	9.888	9.943	10.017
% Recovery	99.6	98.9	99.4	100.2
Std. Deviation	0.089	0.273	0.191	0.112
% RSD**	0.9	2.8	1.9	1.1
Maximum	10.195	10.176	10.176	10.126
Minimum	9.861	9.113	9.436	9.778

* MDL is calculated by multiplying the Standard Deviation by the Student-t value for the number of replicates run.

** %RSD is calculated by dividing the Standard Deviation by the mean, then multiplying by 100.

*** To prepare these standards, refer to Section 9.

15 REVISIONS

April 12, 2019	Rev. 0	AQ400 Method Created and Support Data Collected
April 2020	Rev. 1	Minor editorial changes Added sensitivity and blank absorbance information